



Service Tool Instruction

Service Tool Instruction
Number: 3377855

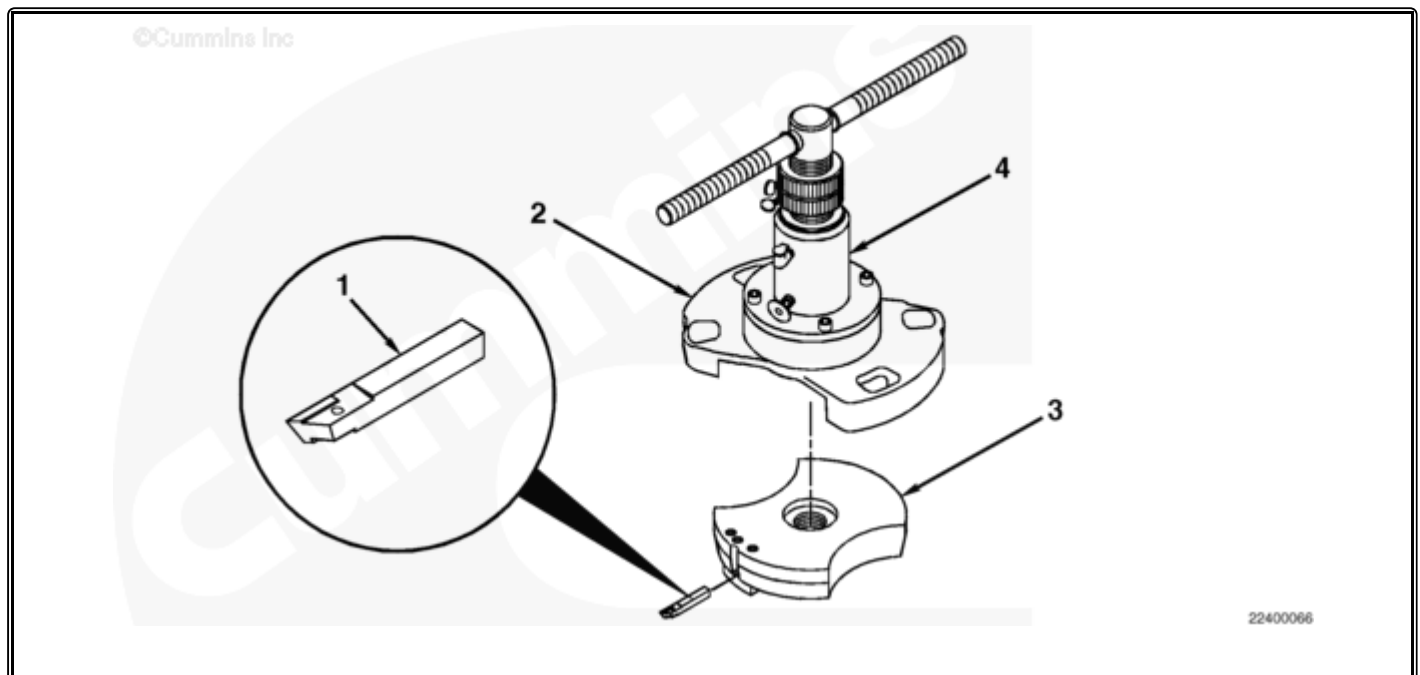
Revision Level:

Released Date: 15-Jul-
2011

Description

K19/QSK19 Counterbore Ledge Chamfering Cutter

Purpose



This service tool instruction provides a procedure for using K19/QSK19 counterbore ledge chamfering cutter, Part Number 3163843. The counterbore tool kit, Part Number 3376563, is required in conjunction with the K19/QSK19 counterbore ledge chamfering cutter. The K19/QSK19 counterbore ledge chamfering tool is designed to cut a 15-degree chamfer on the inside diameter of the counterbore ledge after the cylinder liner lower press fit area has been machined in the field for oversize liners or when the ledge area has been machined due to pitting or other irregularities. Chamfering of the counterbore ledge inside diameter is required **only** if the upper crevice seal liners are to be installed after the block counterbore ledge or lower press fit area has been machined during engine repair/rebuild. If the newly machined bore is **not** machined using this procedure, the upper crevice seal will be damaged when installed with the liner. If standard noncrevice seal liners are to be installed, it is **not** necessary to chamfer the inside diameter of the counterbore ledge. The counterbore ledge inside diameter is machined with a 15-degree chamfer on all production and service cylinder blocks.

Table 1, K19/QSK19 Counterbore Ledge Chamfering Cutter, Part Number 3163843

Item Number	Part Number	Description	Quantity
1	3163843	K19/QSK19 counterbore ledge chamfering cutter	1
Not Shown	3377855	Service Tool Instructions	1

Table 2, Counterbore Tool Kit, Part Number 3376563 (sold separately)

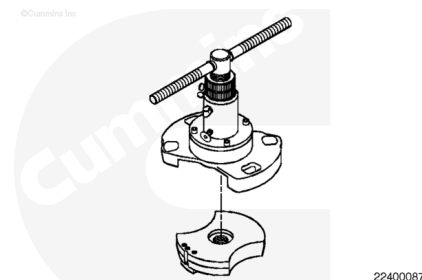
Item Number	Part Number	Description	Quantity
2	3375827	*Base plate	1
3	3824110	*Cutter plate assembly	1
4	3376564	*Drive unit assembly	1

* These parts are required for this procedure and can be purchased as part of the counterbore tool kit, Part Number 3376563, which also includes additional parts **not** shown.

Other shop tools include the following: Depth gauge, feeler gauge, hex key wrench, Dykem™ blue, and shop compressed air.

Assemble/Adjusting the Cutter

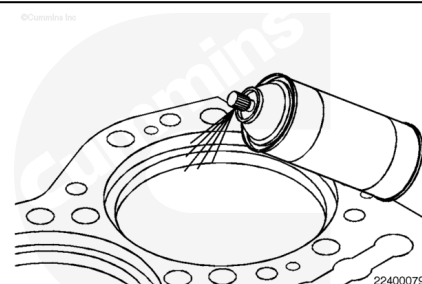
Install the cutter plate onto the drive unit assembly shaft.
Rotate the cutter plate **counterclockwise** until it is hand-tight.



Note : Carefully prepare the cylinder block deck and remove the cylinder head gasket material. The top of the cylinder block deck **must** be clean and free of burrs.

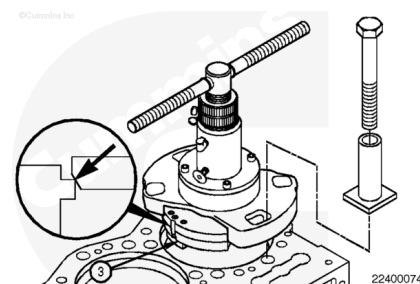
Lightly apply Dykem™ blue over the machined ledge and the counterbore surface.

Allow the surface to dry.



⚠ CAUTION ⚠

The base plate must sit flat on the block surface. Failure to do so can cause the counterbore to be cut incorrectly and will damage the block.

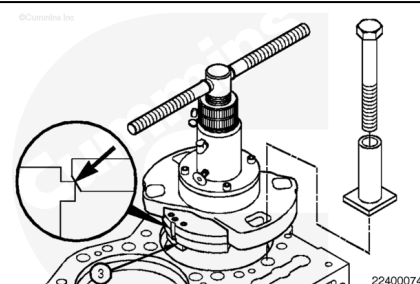


Note : Cutter is not needed for this step.

Lower the cutter plate until it touches the counterbore ledge. The taper on the cutter plate will center the cutter plate and counterbore tool in the cylinder bore. Be sure the base plate does **not** rock on the cylinder block.

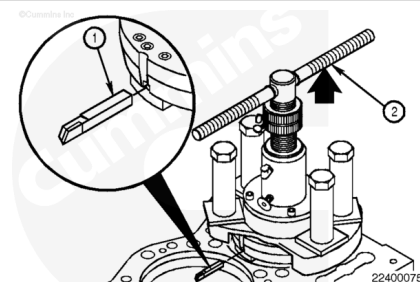
Move the base plate until the capscrew aligns with the cylinder head mounting capscrew locations. Install the four capscrews and four washers.

Torque Value: 45 n·m [33 ft-lb]



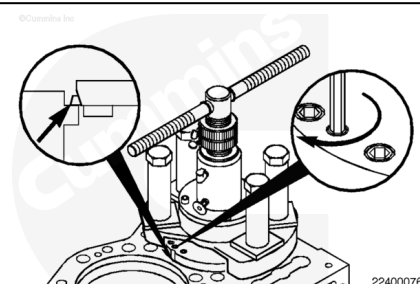
Note : The cutter (1) is first installed upside down. This will locate the cutter relative to the counterbore ledge for the first cut.

Lift the drive unit assembly handle (2) upward, so that the cutter plate comes out of the counterbore. Slide the cutter (1) into the cutter plate.



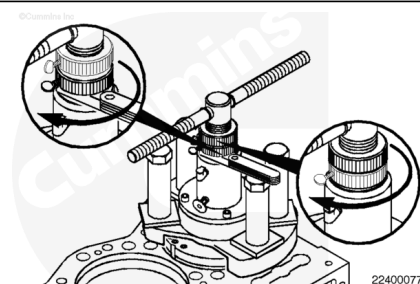
Lower the cutter plate assembly until the cutter rests on the counterbore ledge. Position the cutter in the middle of the counterbore ledge.

Use the 3/32-inch hex key wrench and finger-tighten the setscrew.



Loosen both top and bottom set collars by rotating **counterclockwise**.

Insert a 2 mm [0.080 in] feeler gauge between the drive unit housing and the bottom set collar. Turn the bottom set collar **clockwise** until it touches the feeler gauge. Finger-tighten the bottom collar thumbscrew. Holding the bottom collar in place, rotate the top collar **clockwise** until it touches with the bottom collar. Finger-tighten the top collar thumbscrew.

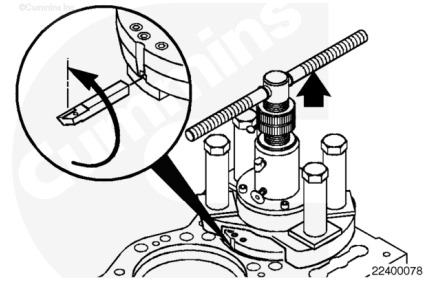


Remove feeler gauge.

Note : The correct position after the cutter has been rotated 180 degrees is shown in the illustration (small hole is facing upward).

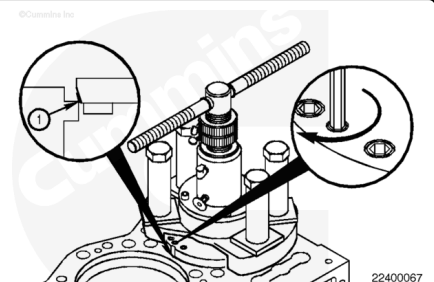
Using a 3/32-inch hex key wrench, loosen the setscrew that is holding the cutter.

Raise upward on the drive unit assembly handle and remove the cutter. Rotate the cutter 180 degrees and then reinstall. Lower the cutter plate into the counterbore; when released, this will drop down 2 mm [0.080 in].



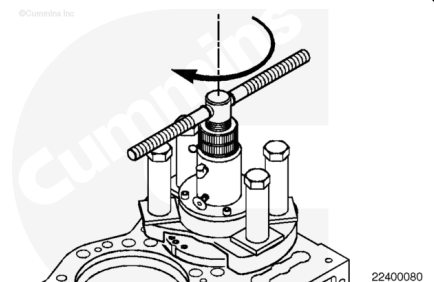
Adjust the cutter (1) outward until its cutting edge is touching the intersection of the ledge and counterbore.

Using a 3/32-inch hex key wrench, rotate the middle setscrew **clockwise** and finger-tighten the setscrew.



Slightly lift the base plate handle upward in preparation to smooth the counterbore ledge surface.

Rotate the base plate handle several times in a **clockwise** direction. This will clean up any burrs or high spots that are on the counterbore ledge.



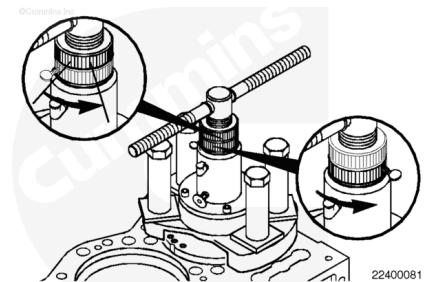
Note : Record the depth of each machining. This figure is used to achieve the correct chamfering cut.

Note : Each mark on the bottom collar is equal to one thousandth of an inch (0.0254 mm [0.001 in]).

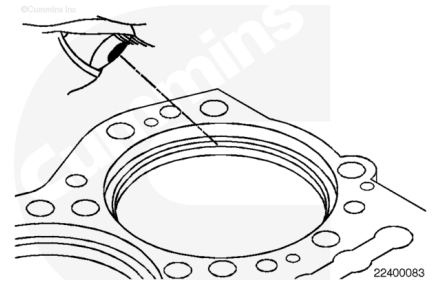
Loosen the top collar thumbscrew. Rotate the top collar **counterclockwise** five marks (0.13 mm [0.005 in]). Finger-tighten the top collar thumbscrew.

Loosen the bottom collar thumbscrew. Rotate the bottom collar **counterclockwise** until it is touching the top collar. Finger-tighten the bottom collar thumbscrew.

Rotate the cutter plate **clockwise** until it advances and completes the cut.

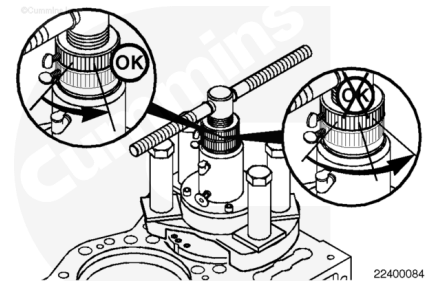


Without removing the cutter assembly from the block, inspect the first cut for correctness of its location.



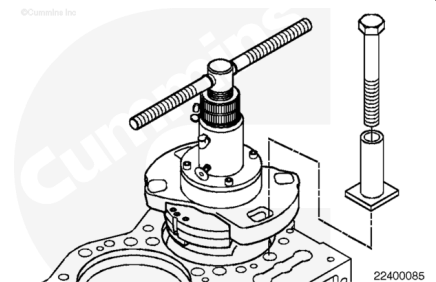
⚠ CAUTION ⚠

Do not remove more than 0.13 mm [0.005 in] per cut. Removing more than this amount can cause the engine block to be unusable.



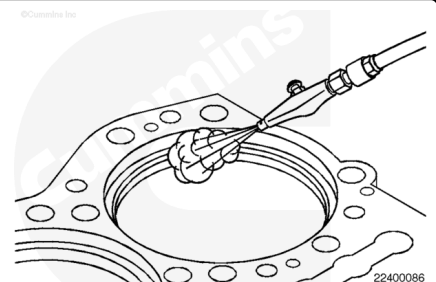
Apply a second cut of 0.13 mm [0.005 in] and inspect. If acceptable, proceed with additional increments of 0.13 mm [0.005 in] cut until the desired depth of chamfer is obtained.

Remove the four capscrews and four washers. Lift the complete drive unit assembly, along with the cutter plate assembly, from the engine block.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

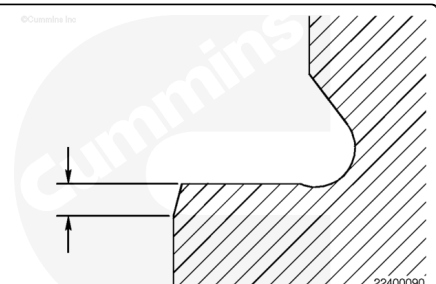


Remove the chips and inspect the counterbore ledge chamfer. Deburr the sharp edges of the newly chamfered counterbore ledge with a crocus cloth.

Measure the depth of chamfer.

Measure from the top of the counterbore ledge cut surface down to the completed cut area.

Vertical length (depth of cut) is 1.00 mm [0.040 in] to 1.25 mm [0.050 in]. Do **not** exceed 1.25 mm [0.050 in] depth of cut.



Last Modified: 04-Oct-2012
